

Lecture 7

Finite Volume methods and Riemann Problem solvers

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Time and space discretisation in 1D

The discretisation process consists of transforming the originally continuous time and space coordinates into discrete variables (see Figure).

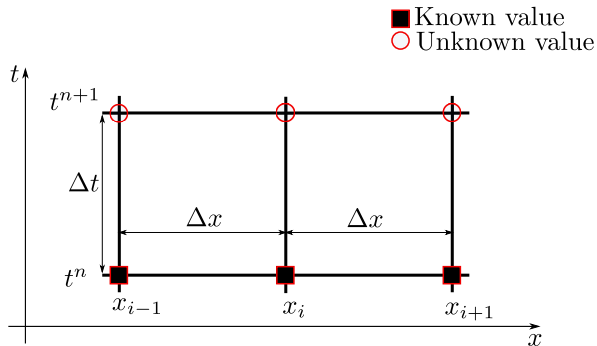


Figure 1: Discretization of time and space in the one-dimensional case.

Time and space discretization in 1D

A set of points (called the calculation points) is defined in time and space by the modeler. The solution is calculated at these points. The governing equations are approximated using the differences between the known and unknown values of the calculation solution at the predefined points. Denoting by U_i^n the solution at the calculation point x_i at the calculation time t^n , the following difference may be seen as a *good* approximation of the derivative $\frac{\partial U}{\partial x}$ over the interval $[x_i, x_{i+1}]$ at the time t^n .

$$\frac{U_i^n - U_{i-1}^n}{x_i - x_{i-1}}$$

This is not the only possible choice. Many alternative formulations may be proposed. The accuracy of the numerical solution depends on the accuracy with which the governing equations are approximated. Approximation methods where the differences between the point values are used to estimate the derivatives are referred to as **finite difference** methods.

In what follows, the distance between points i and $i + 1$ is denoted by Δx . It is often referred to as the grid spacing, or cell width. The difference between two successive calculation times t^n and t^{n+1} , also called the calculation time step, is usually denoted by Δt . The time t^n is usually referred to as *time level* n .

Integral form

Consider the following one-dimensional conservation law:

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} = 0$$

This is called the **differential form** of the conservation law and is valid only for the case in which the solution is smooth throughout. This equation is based on the assumption that the solution is continuous and differentiable with respect to time and space.

In the presence of discontinuities, one must use the **integral form**.

$$\oint [U dx - F(U) dt] = 0 \quad (1)$$

where the line integration is performed along the boundary of the domain in counterclockwise manner.

Integral form of hyperbolic systems

We choose a rectangular control volume V_i in the $x - t$ plane (see the figure), of dimension

$$[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t^n, t^{n+1}]$$

By integrating along the boundary of the control volume:

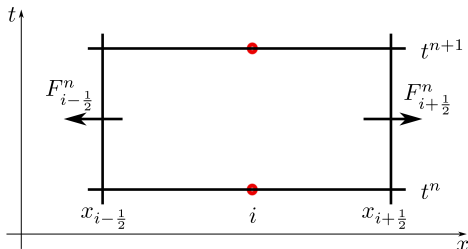


Figure 2: Control volume in the $x - t$ space.

$$\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} [U(x, t^n) dx + \int_{x_{i+\frac{1}{2}}}^{x_{i-\frac{1}{2}}} [U(x, t^{n+1}) dx - \left(\int_{t^n}^{t^{n+1}} [F(U(x_{i+\frac{1}{2}}, t))] dt + \int_{t^{n+1}}^{t^n} [F(U(x_{i-\frac{1}{2}}, t))] dt \right) = 0$$

$$\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} [U(x, t^{n+1}) dx = \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} [U(x, t^n) dx - \left[\int_{t^n}^{t^{n+1}} F(U(x_{i+\frac{1}{2}}, t)) dt - \int_{t^n}^{t^{n+1}} F(U(x_{i-\frac{1}{2}}, t)) dt \right] \quad (2)$$

Derivation of conservative numerical formulation

Let us now define the **mesh size** Δx_i and the **time step** Δt :

$$\Delta x_i = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}, \quad \Delta t = t^{n+1} - t^n$$

We divide through Δx_i and multiply the whole of the second term within the square brackets on the right-hand side by $\Delta t / \Delta t$:

$$\frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} U(x, t^{n+1}) dx = \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} [U(x, t^n) dx - \frac{\Delta t}{\Delta x_i \Delta t} \left[\int_{t^n}^{t^{n+1}} F(U(x_{i+\frac{1}{2}}, t)) dt - \int_{t^n}^{t^{n+1}} F(U(x_{i-\frac{1}{2}}, t)) dt \right)] dx \quad (3)$$

We define integral averages of $U(x, t)$ at times $t = t^{n+1}$ and $t = t^n$ respectively over the length Δx_i (it is a length in the one-dimensional case, in general it is a volume):

$$U_i^{n+1} = \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} U(x, t^{n+1}) dx, \quad U_i^n = \frac{1}{\Delta x_i} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} U(x, t^n) dx \quad (4)$$

We also define time integral averages of the flux $F(U)$ at position $x = x_{i+\frac{1}{2}}$ and $x = x_{i-\frac{1}{2}}$:

$$F_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} F(U(x_{i+\frac{1}{2}}, t)) dt, \quad F_{i-\frac{1}{2}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} F(U(x_{i-\frac{1}{2}}, t)) dt \quad (5)$$

This is the update formula to be used for the numerical integration of the Saint-Venant equations

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x_i} \left[\mathbf{F}_{i+\frac{1}{2}} - \mathbf{F}_{i-\frac{1}{2}} \right] \quad \text{UPDATE FORMULA} \quad (6)$$

N.B.

$\mathbf{U}$ is the vector of unknowns while $\mathbf{F}$ is the vector of fluxes, therefore we have two update formulae, one for each variable, i.e.:

- $$h_i^{n+1} = h_i^n - \frac{\Delta t}{\Delta x} \left[(uh)_{i+\frac{1}{2}} - (uh)_{i-\frac{1}{2}} \right] \quad (7)$$

- $$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} \left[\left(hu^2 + \frac{1}{2}gh^2 \right)_{i+\frac{1}{2}} - \left(hu^2 + \frac{1}{2}gh^2 \right)_{i-\frac{1}{2}} \right] \quad (8)$$

Conservative numerical methods

Substituting the definitions (4) and (5) in the equation (3), we obtain the formula that constitute the basis of **conservative numerical methods**:

$$U_i^{n+1} = U_i^n - \frac{\Delta t}{\Delta x_i} \left[F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}} \right] \quad \text{UPDATE FORMULA} \quad (9)$$

So far the derived expressions are exact expressions and do not involve numerical approximations. However the formula (9) can be interpreted in a numerical sense, if the complete spatial domain is discretized into a set of control volumes $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ called **cells**, $i = 1, 2, \dots, m$. The term $F_{i+\frac{1}{2}}$ is called the inter-cell numerical flux corresponding to the inter-cell boundary at $x = x_{i+\frac{1}{2}}$ between cells i and $i+1$. In general the numerical flux is of the form $F_{i+\frac{1}{2}} = F_{i+\frac{1}{2}}(U_{i-K_L}^n, \dots, U_{i+K_R}^n)$, where the non-negative integers K_L and K_R depend on the particular choice of the numerical flux.

Conservative methods obeys to the so called *telescoping property*. This says that the intercell flux $F_{i+\frac{1}{2}}$ used to update the cell average U_i^{n+1} must be identical to the intercell flux $F_{i-\frac{1}{2}}$ used to update U_{i+1}^{n+1} , so on summation of U_i^{n+1} and U_{i+1}^{n+1} , the flux at the boundary between the cells i and $i+1$ cancels out.

Finite volume approximation

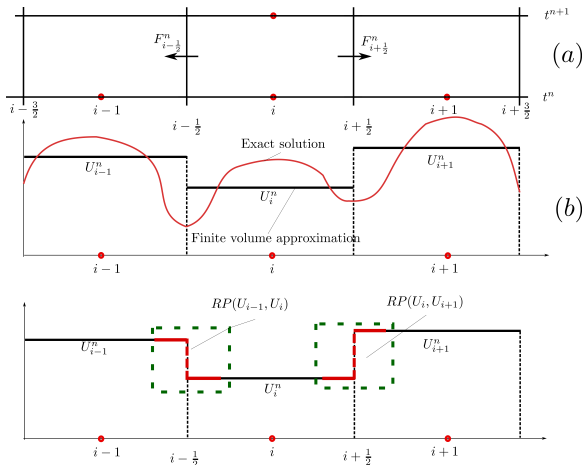


Figure 3: Finite volume approximation and the resulting Riemann problems at interfaces.

Classes of conservative methods

The conservative numerical methods for solving the general initial-boundary value problem for the hyperbolic system are of two distinct classes:

- the **centred or symmetric methods**, that do not explicitly require the provision of wave propagation information to construct the numerical schemes, i.e. they do not require the explicit solution of the Riemann Problem. These schemes are then simple to understand and to implement, particularly for complicated hyperbolic system, as for example the Saint-Venant-Exner model.
- the **upwind methods**, or Godunov-type methods, that use the wave propagation information, essential property of the hyperbolic partial differential equations, to construct the numerical schemes. This is achieved in various ways. At the highest level one solves local Riemann problem exactly; at the lowest level one provides a minimum of information on wave propagation directions, perhaps just the sign of a single wave at each intercell boundary.

Centred methods: Lax-Friedrichs

Here we present schemes that do not require the (explicit) solution of the Riemann problem. These schemes are not biased by the wave propagation direction, which distinguishes upwind methods, and are called centred or symmetric schemes.

Numerical schemes may be obtained from the conservative formula(9) by giving appropriate definitions for the inter-cell flux $\mathbf{F}_{i+\frac{1}{2}}$. The Lax-Friedrichs method results if we choose:

$$\mathbf{F}_{i+\frac{1}{2}}^{LF} = \frac{1}{2} [\mathbf{F}(\mathbf{U}_{i+1}^n) + \mathbf{F}(\mathbf{U}_i^n)] - \frac{1}{2} \frac{\Delta x}{\Delta t} (\mathbf{U}_{i+1}^n - \mathbf{U}_i^n) \quad (10)$$

This **first-order accurate** scheme is exceedingly simple to implement but, as is well known, is too diffusive to be used in practical computations. The scheme is **monotone** and has linearised stability condition:

$$\Delta t \leq \frac{CFL \Delta x}{|u| + \sqrt{gh}}$$

with $CFL = 0.9$ in practical computations.

Centred methods: Lax-Wendroff

The Lax-Wendroff method results if we choose:

$$\mathbf{F}_{i+\frac{1}{2}}^{LW} = \mathbf{F} \left(\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} \right), \quad (11)$$

with

$$\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} = \frac{1}{2}(\mathbf{U}_i^n + \mathbf{U}_{i+1}^n) - \frac{1}{2} \frac{\Delta t}{\Delta x} [\mathbf{F}(\mathbf{U}_{i+1}^n) - \mathbf{F}(\mathbf{U}_i^n)], \quad (12)$$

The two-step Lax-Wendroff scheme is not monotone!!! This scheme is **second-order accurate** and **is not monotone** and thus produces spurious oscillations in the vicinity of high gradients, such as in shock waves. It has linearised stability condition:

$$\Delta t \leq \frac{CFL \Delta x}{|u| + \sqrt{gh}}$$

with $CFL = 0.9$ in practical computations.

Centred methods: FORCE

Another possible choice of flux is that of the force scheme (First ORder CEntred) proposed by Toro in 2001; this may be written as:

$$\mathbf{F}_{i+\frac{1}{2}}^{FORCE} = \frac{1}{2} \left(\mathbf{F}_{i+\frac{1}{2}}^{LF} + \mathbf{F}_{i+\frac{1}{2}}^{LW} \right), \quad (13)$$

The force scheme is **first-order accurate** and has half the numerical viscosity of the Lax-Friedrichs method. The scheme has been proved to be **monotone and stable**, with linearised stability condition

$$\Delta t \leq \frac{CFL \Delta x}{|u| + \sqrt{gh}}$$

with $CFL = 0.9$ in practical computations. The FORCE scheme can be written in a two-step staggered grid version as

$$\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} = \frac{1}{2} (\mathbf{U}_i^n + \mathbf{U}_{i+1}^n) - \frac{1}{2} \frac{\Delta t}{\Delta x} [\mathbf{F}(\mathbf{U}_{i+1}^n) - \mathbf{F}(\mathbf{U}_i^n)], \quad (14)$$

$$\mathbf{U}_i^{n+1} = \frac{1}{2} \left(\mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}} + \mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \frac{1}{2} \frac{\Delta t}{\Delta x} \left[\mathbf{F}(\mathbf{U}_{i+\frac{1}{2}}^{n+\frac{1}{2}}) - \mathbf{F}(\mathbf{U}_{i-\frac{1}{2}}^{n+\frac{1}{2}}) \right], \quad (15)$$

The Godunov upwind method

The upwind method of Godunov has been proposed for the first time in 1959 [1]. It utilises the solution of the Riemann problem locally; this solution can be exact or approximate as we will see.

- [1] S.K. Godunov (1959). Finite difference methods for the computation of discontinuous solution of the equations of fluid dynamics. *Mat. Sb.*,47:271-306

The Godunov upwind method

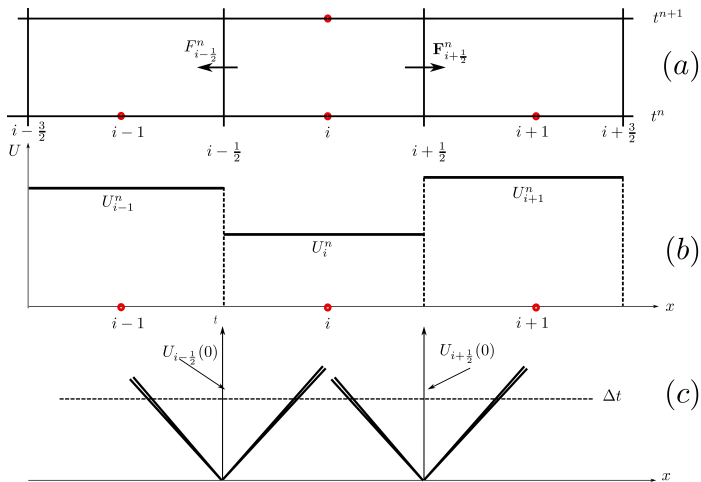


Figure 4: Godunov upwind method: (a) control volume in $x - t$ space, (b) integral averages give piece-wise constant data U_{i-1} , U_i , U_{i+1} , (c) structure of solutions of Riemann problems at intercell boundaries

The Godunov upwind method: the scheme

Let us assume that the initial data U^n at time $t = t^n$ is a set of integral averages U_i^n over control volumes $I_i = x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}$ and this results in a piece-wise constant distributions of data. Figure (10(b)) shows a possible distribution of the data in cells $i-1$, i and $i+1$ for a typical component of the vector U^n .

Let us consider a short portion of the domain, say $x_{i-\frac{1}{2}} \leq x \leq x_{i+\frac{3}{2}}$, then locally one has the set of conservation laws with initial data that consists of two constant states separated by a discontinuity. That is, we have the following initial value problem (IVP):

$$\left. \begin{aligned} \partial_t U + \partial_x F &= \mathbf{0}, \quad x \in \mathcal{R}, \quad t > 0, \\ U(x, t^n) &= \begin{cases} U_i & \text{if } x < x_{i+\frac{1}{2}}, \\ U_{i+1} & \text{if } x > x_{i+\frac{1}{2}}. \end{cases} \end{aligned} \right\} \quad (16)$$

This is precisely a Riemann problem, whose data states are U_i (left) and U_{i+1} (right), the solution of which is denoted by $U_{i+\frac{1}{2}}(x, t)$.

The Godunov upwind method: the scheme

Let us consider the other portion of the domain, say $x_{i-\frac{3}{2}} \leq x \leq x_{i+\frac{1}{2}}$, then we have another initial value problem (IVP):

$$\left. \begin{aligned} \partial_t U + \partial_x F &= \mathbf{0}, \quad x \in \mathcal{R}, \quad t > 0, \\ U(x, t^n) &= \begin{cases} U_{i-1} & \text{if } x < x_{i-\frac{1}{2}}, \\ U_i & \text{if } x > x_{i-\frac{1}{2}}. \end{cases} \end{aligned} \right\} \quad (17)$$

This is precisely a Riemann problem, whose data states are U_{i-1} (left) and U_i (right), the solution of which is denoted by $U_{i-\frac{1}{2}}(x, t)$.

The Godunov upwind method: the scheme

The Godunov flux $F_{i+\frac{1}{2}}$ at the intercell boundary $x_{i+\frac{1}{2}}$ may be defined as the physical flux function $F(U)$ evaluated at the solution $U_{i+\frac{1}{2}}(x, t)$, which in turns is to be evaluated along the t -axis; in local coordinates this is $x/t = 0$. Then, we can write

$$F_{i+\frac{1}{2}} = F\left(U_{i+\frac{1}{2}}(0)\right) \quad (18)$$

Two items are needed to evaluate the Godunov flux

- the solution $U_{i+\frac{1}{2}}(x, t)$ of the Riemann problem with data $U_L \equiv U_i^n$ (left) and $U_R \equiv U_{i+1}^n$ (right);
- a solution sampling procedure to correctly identify the required value along the t -axis, x/t .

Dam-break problem

In the language of free surface water flows, a rarefaction is also called a depression and a shock is also called a bore. Here we shall use both terminologies. If one assumes that the wave phenomenon is correctly governed by the Saint-Venant equations and that the wall vanishes instantaneously at time $t = 0$, then the situation may be described by Figs (5)-(7). The initial conditions for the water depth at time $t = 0$ are represented by Fig. (5) where the wall is replaced by a discontinuity in water depth at the position $x = 25$ m. Fig. (5) shows the water depth, the velocity profile and the a summary of the wave process as a function of space and time at the later time $t = 1$ s after the wall collapses, while the situation at $t = 2$ s and $t = 3$ s is illustrated in Fig. (6) and (7) respectively. The right wave is a shock, a discontinuous wave, and the left wave is a rarefaction, a smooth wave.

Dam-break problem

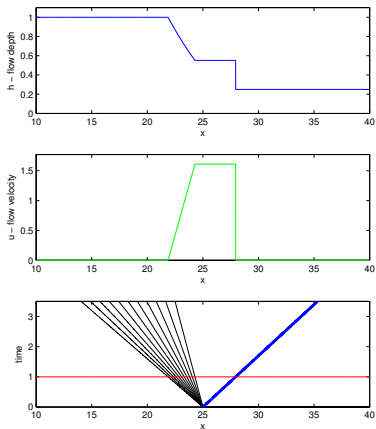


Figure 5: Time evolution of the dam-break problem: $t=1$ s

Dam-break problem

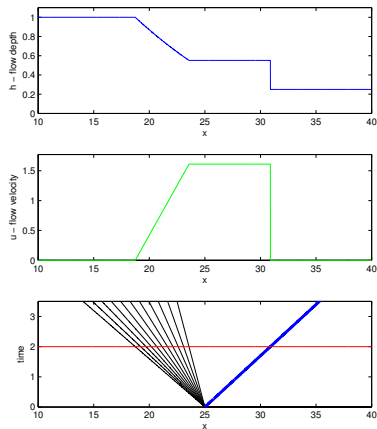


Figure 6: Time evolution of the dam-break problem: $t = 2s$

Dam-break problem

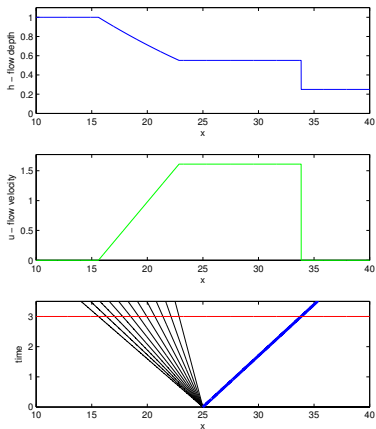


Figure 7: Time evolution of the dam-break problem: $t = 3\text{s}$

The Riemann Problem

The Riemann problem for the Saint-Venant equations is a generalisation of the dam-break problem. Formally, the Riemann problem is defined as the initial-value problem (IVP)

$$\left. \begin{aligned} \partial_t \mathbf{U} + \partial_x \mathbf{F} &= \mathbf{0}, \quad x \in \mathcal{R}, \quad t > 0, \\ \mathbf{U}(x, 0) &= \begin{cases} \mathbf{U}_L & \text{if } x < 0, \\ \mathbf{U}_R & \text{if } x > 0. \end{cases} \end{aligned} \right\} \quad (19)$$

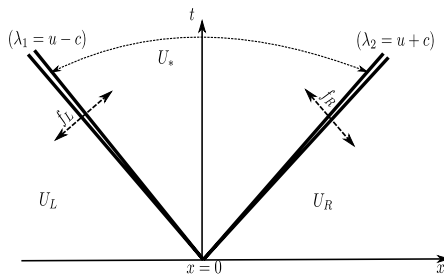


Figure 8: Structure of the solution of the Riemann problem (19) .

The Riemann Problem

The structure of the similarity solution of the problem is shown below in the entire x - t half plane. There are two wave families. The left family is associated with the eigenvalue λ_1 , while the right wave family is associated with λ_2 . Waves associated with the *genuinely non-linear* characteristic fields λ_1 and λ_2 are either shocks (discontinuous solutions) or rarefactions (smooth solutions). The entire solution consists of three constant states, namely $\mathbf{Q}_L$ (data), $\mathbf{Q}_*$, and $\mathbf{Q}_R$ (data), separated by two waves. The unknown states to be found are $\mathbf{Q}_*$. If any of the λ_1 and λ_2 waves is a rarefaction then there will be a smooth transition between two adjacent constant states. In order to solve exactly the entire initial-value problem we need to establish appropriate jump conditions across each characteristic field to connect the unknown state $\mathbf{Q}_*$ to the initial conditions $\mathbf{Q}_L$ (left) and $\mathbf{Q}_R$ (right) respectively. In what follows we establish such jump conditions across each characteristic field.

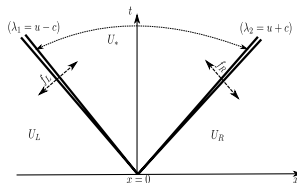


Figure 9: Structure of the solution of the Riemann problem (19).

The Riemann Problem

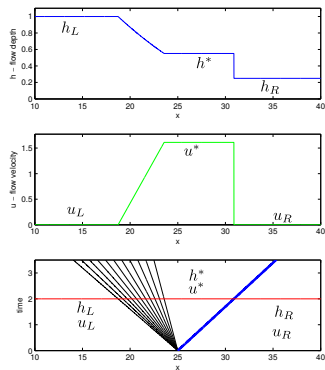


Figure 10: Structure of the solution of the Riemann problem: solution in the star region.

The Riemann Problem

Why is relevant the study of the Riemann Problem?

- Because we can build exact solutions to compare with numerical ones;
- because it is usually used for building numerical methods.

Exact solution

The exact solution of the Riemann problem in the star region can be determined through the application of appropriate *jump* conditions through the waves, either shock or rarefactions. Through shock waves the Rankine-Hugoniot conditions holds, whereas through rarefaction waves, characteristic equations can be impose.

Details on the exact solution of the Riemann problem for the Saint-Venant equations can be found in the text book by Toro, E.F. *Shock-capturing methods for free-surface shallow flows*. Wiley and Sons Ltd; 2001.

The provided Matlab codes solve a suite of Riemann problems for the Saint-Venant equations (1D problem) via both approximate and exact solutions.

Approximate Riemann solvers

To compute numerical solutions by Godunov-type methods, one can use the exact or approximate Riemann solvers. Approximate solvers, if used judiciously, can provide effective computational tools at a competitive cost. Making the choice between the exact and approximate Riemann solvers is motivated by

- computational cost;
- simplicity;
- correctness.

Correctness should be the overriding criterion.

Approximate Riemann solvers

In this approach one first computes an approximate solution $\mathbf{U}_{i+\frac{1}{2}}(x/t)$ to the Riemann problem. then an approximate numerical flux is obtained by evaluating the physical flux

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}(\mathbf{U}_{i+\frac{1}{2}}(0)) \quad (20)$$

where $\mathbf{U}_{i+\frac{1}{2}}(0)$ is the appropriate value along the t -axis. The process of finding the approximate solution $\mathbf{U}_{i+\frac{1}{2}}(0)$ has two steps. In the first step one computes an approximate solution for the variable h^* and u^* in the star region. In the second step one samples the solution to obtain the correct values along the t -axis and evaluate the Godunov flux.

The HLL (Harten, Lax, van Leer) Riemann solver

If we apply similar considerations to the left and right portion of the Riemann fan, we obtain the following relations for the fluxes:

$$\mathbf{F}^{HLL} = \mathbf{F}_L + S_L(\mathbf{U}^{HLL} - \mathbf{U}_L) \quad (21)$$

or

$$\mathbf{F}^{HLL} = \mathbf{F}_R + S_R(\mathbf{U}^{HLL} - \mathbf{U}_R) \quad (22)$$

and finally we get the HLL flux:

$$\mathbf{F}^{HLL} = \frac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} \quad (23)$$

The corresponding HLL intercell flux for the approximate Godunov method is then given by

$$\mathbf{F}_{i+\frac{1}{2}}^{HLL} = \left\{ \begin{array}{lll} \mathbf{F}_L & \text{if} & 0 \leq S_L, \\ \mathbf{F}^{HLL} & \text{if} & S_L \leq 0 \leq S_R, \\ \mathbf{F}_R & \text{if} & 0 \geq S_R. \end{array} \right\} \quad (24)$$

The HLL Riemann solvers

As to the wave speed estimates S_L and S_R there are several possible choices available. We suggest the following:

$$S_L = u_L - c_L g_K \quad S_R = u_R + c_R g_K \quad (25)$$

where g_K ($K = L, R$) is given by

$$g_K = \left\{ \begin{array}{ll} \sqrt{\frac{(h^* + h_K)h^*}{2h_K^2}} & \text{if } h^* > h_K, \\ 1 & \text{if } h^* \leq h_K. \end{array} \right\} \quad (26)$$

here h^* is an estimate for the exact solution for h in the star region. A successful choice for h^* is:

$$h^* = \frac{1}{g} \left[\frac{1}{4}(u_L - u_R) + \frac{1}{2}(c_L + c_R) \right]^2, \quad (27)$$

where $C_L = \sqrt{gh_L}$ and $C_R = \sqrt{gh_R}$.

Exercise

The Folder *Saint_Venant_FV_FD* contains a Matlab code which solves the Saint-Venant equations (1D problem) using 5 different Finite Volumes and 3 Finite Differences methods. The domain is discretized using equally grid spacing Δx (dx in the code). The initial condition is a Riemann problem (test 1 and test 2). The numerical solution (blue line-dots) is compared with the exact solution (black solid-line).

- ① Which are the main differences between the centred (Lax-Friedrichs, Lax-Wendroff, FORCE) and Godunov methods (Exact solution of RP, HLL)? Which solutions are most accurate?
- ② Which are the main differences between the Finite Volume and Finite Difference methods? Which solutions are most accurate?
- ③ How the numerical solution behaves if the grid spacing is reduced? Which are the consequences in terms of time computation?
- ④ What does happen if the CFL is > 1 ?

Refence Textbooks

- 1 Guinot, V. Wave propagation in Fluids: models and numerical techniques, Wiley and Sons Ltd; 2010.
- 2 Toro, E.F. Shock-capturing methods for free-surface shallow flows. Wiley and Sons Ltd; 2001.
- 3 Toro, E.F. Riemann solvers and numerical methods for fluid dynamics, Springer Verlag; Third Edition, 2009.